

# CONFIDENT-TAIL TEST-TIME ADAPTATION MOVES THE THRESHOLD, NOT THE RANKING, IN AUDIO DEEPPAKE DETECTION

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## ABSTRACT

Cross-corpus failure in audio deepfake detection is often a failure of the decision threshold rather than of score ordering: a released detector in our In-the-Wild evaluation has ROC-AUC 0.9171 but 39.08% accuracy at threshold 0.5. Confident-tail self-training, which pseudo-labels the lowest- and highest-scoring target clips, exploits the ordering that survives. Over 79 adaptation runs spanning four released checkpoints, two corpora, spoof prevalence 0.01 to 0.972 and budgets 0.02 to 0.50, it moves the decision far more than the ordering that decision is applied to: accuracy at threshold 0.5 shifts by a median of 15.1 points while EER shifts by 0.38 and ROC-AUC by 0.12. Where the decision lands follows a counting quantity. A bucket assigned class  $c$  with realized fraction  $b_c$  of a pool with prior  $\pi_c$  cannot be purer than  $\min(1, \pi_c/b_c)$ , so a fraction  $\Delta = \sum_c (b_c - \pi_c)^+$  of the pool is necessarily mislabeled. Pooled across checkpoints and priors,  $\Delta$  orders the resulting threshold gap where the budget alone does not (Spearman +0.75 against +0.36). Purity orders none of it, and neither does the budget:  $q = 0.50$  helps at 37% spoof prevalence while  $q = 0.30$  destroys accuracy at 97%, and on a balanced pool a perfectly pure selection still costs  $9.6 \pm 0.8$  points. The rule chooses an operating point; it does not improve a detector.

**Index Terms**— audio deepfake detection, test-time adaptation, label shift, pseudo-labels

## 1. INTRODUCTION

Self-supervised speech representations paired with anti-spoofing back-ends provide strong audio deepfake detectors [1, 2], but transfer across recording conditions and synthesis systems remains difficult [3]. A target-domain failure can lie in the score ordering, in the decision threshold, or in both: a released detector in our In-the-Wild evaluation has ROC-AUC 0.9171 but 39.08% accuracy at threshold 0.5.

Test-time adaptation (TTA) updates a source model from unlabeled target observations, and both entropy minimization [4] and pseudo-label self-training can reinforce incorrect predictions. We examine one self-training rule, assigning the

bottom and top score quantiles to bona fide and spoof, and ask what it actually changes. It is not a rule proposed elsewhere: it is our own prior design, an instance of confidence-based pseudo-label selection [5, 16] with a fixed budget per tail. We find it changes the operating point and little else, and that a counting constraint on the budget predicts where that point lands.

Class imbalance in adaptation is established ground: Label Shift Adapter [6] adapts parameters to estimated target proportions, RLSbench [7] studies joint label and conditional shift, black-box shift estimation (BBSE) [8] recovers target priors from a source confusion matrix, and T<sup>2</sup>A [9] treats deepfake TTA by uncertainty-aware negative learning. Our claim is narrower and empirical: what this rule changes, and the counting quantity that predicts it.

## 2. TAIL SELECTION AND ADAPTATION

Let  $y = 1$  denote spoof,  $s_\theta(x) \in [0, 1]$  the detector’s spoof score, and the unlabeled adaptation pool have  $N$  clips with unknown class fractions  $\pi_c$ . Target labels are used only for evaluation, controlled resampling and explicitly marked oracle analyses.

### 2.1. A necessary condition for pure pseudo-labels

A bucket  $C_c$  assigned pseudo-label  $c$ , with realized fraction  $b_c = |C_c|/N > 0$ , contains at most  $\pi_c N$  correct examples. Thus

$$\text{purity}(C_c) \leq \min\{1, \pi_c/b_c\}. \quad (1)$$

This is a finite-pool counting bound and makes no assumption about the model. An update, however, does not consume a purity: it consumes wrongly labeled examples. The same counting fact says that the selection necessarily mislabels a fraction

$$\Delta = \sum_c (b_c - \pi_c)^+ \quad (2)$$

of the pool. Purity is a within-bucket rate and  $\Delta$  a share of the pool, and the two order configurations differently: a small bucket can have very low purity while mislabeling almost

nothing. Section 4 finds that  $\Delta$  orders the observed outcomes and that purity does not.

For exact symmetric tails,  $b_0 = b_1 = q$ , both statements reduce to  $\Delta = (q - \min\{\pi, 1 - \pi\})^+$ , so perfect purity is possible only if

$$q \leq \min\{\pi, 1 - \pi\}, \quad \pi = \pi_1. \quad (3)$$

At 97.22% spoof prevalence a bucket labeling 30% of the pool bona fide has purity at most 9.27%, and no amount of AUC removes that. Feasibility is not sufficiency: score overlap contaminates feasible tails, and learning tolerates impure labels when there are few of them. That is why we test (2) rather than (3).

A prior allocation uses nominal fractions  $q_c = 2q\hat{\pi}_c$ , keeping the total selection fraction  $2q$ . With exact buckets and a correct prior,  $q \leq 1/2$  ensures  $q_c \leq \pi_c$ ; with estimated priors it instead requires  $\hat{\pi}_c/\pi_c \leq 1/(2q)$ , a limit on overestimation rather than a safety theorem. A smaller symmetric  $q$  needs no point estimate, but a guaranteed feasible nonzero value still needs a lower bound on the minority prevalence.

**Quantile ties.** Inclusive thresholds and spoof precedence on overlap make realized tail sizes differ from nominal budgets, so (1) and (2) are evaluated at the realized  $b_c$  throughout.

## 2.2. Prior estimation and objective

For  $M_{ij} = P_S(\hat{y} = j \mid y = i)$  and a target prediction histogram  $h$  from source predictions on the adaptation pool, BBSE solves  $M^\top \hat{p} = h$ . We estimate  $M$  from up to 4,000 class-balanced clips of the checkpoint’s training corpus, clip to  $[10^{-4}, 1 - 10^{-4}]$  and renormalize; using training rather than held-out data makes its error rates optimistic.

At each of four passes, the current model scores the adaptation pool and refreshes pseudo-labels. For a mini-batch  $B$  and its selected subset  $B_C$ ,

$$\begin{aligned} \mathcal{L}_B = & \frac{1}{|B_C|} \sum_{x \in B_C} \text{CE}(p_\theta(x), \tilde{y}_x) \\ & + \frac{\lambda}{2|B|} \sum_{x \in B} \|p_\theta(a(x)) - \text{sg}[p_\theta(x)]\|_2^2, \end{aligned} \quad (4)$$

where  $p_\theta = (1 - s_\theta, s_\theta)$ , the CE term is omitted when  $B_C$  is empty, and  $\lambda = 0.3$ . The consistency term uses all batch examples and both probability components. The perturbation is per-clip gain sampled uniformly from  $[0.7, 1.3]$  plus Gaussian noise with standard deviation 0.005. Only the last four transformer blocks’ LayerNorm affine parameters and the classifier head are updated; BatchNorm statistics stay frozen. Adam uses learning rate  $10^{-4}$  and batch size 32. The released DeepFense models have 16,706 trainable parameters under this rule. A fixed-confidence control instead selects scores at most 0.05 or at least 0.95, with the same objective.

**Table 1.** The 400,435 available DF eval trials, DF-2, adaptation seed 0, whose always-spoof accuracy baseline is 97.22%. Threshold 0.5 except in the two frozen-source controls. All columns but AUC are percentages;  $G_\tau$  is an oracle diagnostic and smaller is better.

| Method                | EER  | AUC    | Acc.  | BA    | $G_\tau$ |
|-----------------------|------|--------|-------|-------|----------|
| Source                | 4.51 | 0.9923 | 98.87 | 88.28 | 0.20     |
| Entropy (Tent-style)  | 4.28 | 0.9765 | 98.52 | 73.51 | 0.57     |
| Fixed confidence 0.95 | 4.30 | 0.9918 | 99.02 | 85.03 | 0.00     |
| Symmetric $q = 0.02$  | 4.19 | 0.9930 | 98.96 | 83.88 | 0.01     |
| Symmetric $q = 0.3$   | 5.84 | 0.9846 | 62.96 | 80.89 | 35.38    |
| BBSE tail budget      | 4.06 | 0.9931 | 98.95 | 82.34 | 0.02     |

## 3. EVALUATION PROTOCOL

**Checkpoints and pools.** We use DeepFense XLS-R 300M + AASIST checkpoints trained on ASVspoof2019-LA with seeds 2, 42 and 240 [10, 11], denoted DF-2, DF-42, DF-240, and a WaveFake-trained SSL-AASIST checkpoint WF [12]. Audio is 16 kHz, cropped to 64,000 samples for DeepFense and 64,600 for WF.

The official ASVspoof2021-DF metadata has 533,928 eval trials among 611,829 across three phases [13]. Our available *eval-only* subset holds 400,435 of them, of which 389,299 are spoof and 11,136 bona fide, so 97.219% spoof, covering 75.00% of the spoof class and 74.89% of bona fide. The missing shard rules out a full-set comparison; all comparisons use the same trials.

The main DF run adapts on 20,000 clips with adaptation seed 0; we report the available pool and its 380,435-clip disjoint remainder. Seeds 1 and 2 adapt 20,000 clips of a fixed 50,000-trial subset (seed 12345), sensitivity checks on overlapping data rather than nine independent datasets. In-the-Wild uses all 31,779 clips (11,816 spoof, 37.182%) with 20,000 adaptation clips.

**Prevalence intervention.** We resample either corpus to a target spoof proportion (seed 4242) by thinning whichever class must shrink: In-the-Wild to eight priors from 0.01 to 0.97 including its native 0.372, and DF to 0.01, 0.10, 0.50 and 0.90, the last a balanced pool. Every resampled pool adapts on 8,000 clips, so the optimization budget is identical at every prevalence and a disjoint remainder always remains; the evaluation pool size still varies with prevalence, which the experiment does not separate from it.

**Metrics and controls.** Metrics are recomputed from saved float32 probabilities. ROC-AUC gives ties half credit; EER interpolates the FPR/FNR crossing with identical scores grouped, and need not correspond to an attainable deterministic threshold. We also report accuracy, balanced accuracy, and the oracle threshold gap  $G_\tau = \max_t \text{Acc}(s, t) - \text{Acc}(s, \tau)$ , whose maximum ranges over all distinct thresholds including constant predictions. It uses evaluation labels and is a diagnostic of threshold headroom, not a deployable method or a calibration error.

Two label-free controls keep source scores fixed and pick

a threshold from adaptation scores: their median, or their  $(1 - \hat{\pi})$ -quantile using BBSE. Both preserve AUC and EER; neither is a claim of Bayes optimality.

We additionally report fixed-configuration entropy adaptation (Tent-style) under the same update settings [4].<sup>1</sup>

## 4. RESULTS

### 4.1. Adaptation moves the decision, not the ranking

Across all 79 symmetric-budget runs – four checkpoints, two corpora, spoof prevalence 0.01 to 0.972, budgets  $q$  from 0.02 to 0.50 and three adaptation seeds on eight cells – adaptation changes where the decision falls far more than it changes the ordering that decision is applied to. Accuracy at threshold 0.5 moves by a median of 15.14 points, with a 90th percentile of 31.69 and a maximum of 43.31, while EER moves by 0.38 (1.33, 4.00) and ROC-AUC by 0.12 points (0.67, 3.02). Accuracy moves more than ten times as far as EER in 66 of the 79 runs and further than EER in 78 of them (Fig. 1(a)); the same DF-2 run that changes accuracy by 35.9 points changes EER by 1.33.

EER and AUC are threshold-free and neither is pinned by the class prior, so this comparison does not rest on decomposing accuracy into a ranking-determined ceiling and a threshold gap  $G_\tau$ . We use that decomposition below only as bookkeeping, and it is not independent evidence: the identity  $\text{Acc}(\tau) = \max_t \text{Acc}(t) - G_\tau$  makes “accuracy improves whenever  $G_\tau$  falls” automatic once the ceiling term is small, and at skewed prevalence that term is weakly identified, sitting 1.7 to 3.0 points above the always-majority rule on average and within one point of it in 26 of the 79 runs.

### 4.2. Contamination sets where the operating point lands

The quantity that predicts where the decision lands is  $\Delta$ . Within one checkpoint and prevalence  $\Delta$  is a monotone transform of  $q$ , so a within-stratum correlation cannot separate the two: the 12 of 14 such strata in which the post-adaptation gap rises with  $\Delta$  establish only that larger budgets place the decision worse. The question is whether  $\Delta$  carries across

strata. Pooled over all 63 single-seed cells it does, at Spearman  $\rho = +0.75$  against  $+0.36$  for  $q$  and  $-0.47$  for the minority prevalence alone (Fig. 1(b)). A matched-pair test is sharper: across pairs drawn from different checkpoints or prevalences that agree on  $\Delta$  to within 0.02 but differ in  $q$  by more than 0.10, the post-adaptation gaps differ by a median of 7.8 points, where pairs agreeing on  $q$  but differing in  $\Delta$  by more than 0.10 differ by 28.9. Contamination pins the outcome about four times as tightly as the budget does, though 7.8 points of residual spread shows it is not the whole story.

Neither purity nor the budget orders these outcomes. At the native In-the-Wild prior  $q = 0.50$  – the largest symmetric budget there is, labeling the entire pool – leaves a 7.43-point gap and gains 7.47 points of balanced accuracy, while  $q = 0.15$  at  $\pi = 0.01$  leaves 28.17 and gains 2.64; a tail of purity 0.107 gains 9.36 points where one of purity 0.320 loses 0.37. Nor is a clean selection sufficient: on a balanced DF pool, where  $\Delta = 0$  and the measured purity is 1.000, adaptation still moves the gap from 7.38 to  $17.4 \pm 0.9$  and costs  $9.6 \pm 0.8$  points of accuracy across three seeds.

### 4.3. Anchoring on ASVspoof2021-DF

On DF-2 (Table 1) symmetric  $q = 0.3$  raises EER from 4.51% to 5.84%, lowers balanced accuracy from 88.28% to 80.89% and accuracy from 98.87% to 62.96%, against a 97.22% always-spoof baseline and a 35.38-point threshold gap; a frozen-source BBSE threshold alone reproduces that accuracy, at 98.88%. Allocating the tails by a BBSE prior estimate reaches 4.06% EER and 98.95% accuracy but 82.34% balanced accuracy, below the source: one effective remedy, not a uniformly best one.

The same pattern holds on DF-42 and DF-240. Across the nine checkpoint/run combinations BBSE has lower EER than the symmetric budget in 9/9, and than the source in 8/9 on the available pool and 7/9 under disjoint evaluation. A paired stratified bootstrap over trials (2,000 replicates, resampled within class) separates the two: all nine symmetric differences have 95% intervals excluding zero, but only 4/9 and 5/9 of the source differences do, and DF-240’s is  $-0.02$  points  $[-0.14, +0.12]$ . These counts are descriptive, not independent evidence.

### 4.4. Corpus shift and prior estimation

At the native In-the-Wild prior the budget has no counting obstruction and improves accuracy on three of four checkpoints. For DF-42 it raises AUC from 0.9171 to 0.9473 and accuracy from 39.08% to 82.39%, though a frozen-source median threshold already reaches 78.10%. BBSE estimates 97.9% spoof on this 37.2%-spoof corpus and lands at 14.54% EER and 47.99% accuracy, against 11.59%/91.28% for a non-deployable oracle. BBSE is identifiable under label shift with an invertible confusion matrix [8], but under unrestricted conditional shift  $P_{\mathcal{T}}(\hat{y} = 1) = \pi \text{TPR}_{\mathcal{T}} + (1 - \pi) \text{FPR}_{\mathcal{T}}$  cannot

<sup>1</sup>Its scores saturate: 98.07% lie within  $10^{-6}$  of 1.0 but only 93.02% exactly at 1.0, and conflating the two makes its AUC look impossible. With exact ties grouped it reaches 4.28% EER and 0.9765 AUC against the source’s 4.51% and 0.9923, a gain near one operating point coexisting with worse integrated ranking. The scoring convention matters too: ROC interpolation gives the ETA ablation 12.31% EER where a nearest-point rule gives 7.44%. Three further published TTA methods were run the same way, as controlled variants rather than tuned reproductions: an ETA reliability-only ablation without redundancy or Fisher terms (EER 12.31%, and an all-spoof decision at threshold 0.5 on DF-42 and DF-240), SAR with recovery threshold scaled to  $(0.2/\ln 1000) \ln 2$  (4.48%), and IM-PL, a SHOT-inspired information-maximization control with probability-space centroid pseudo-labels that freezes the head and updates 16,384 parameters (6.35%) [14–16]. None is a tuned reproduction and none establishes anything about those methods’ published performance; their audited rows are in the released tables.

**Table 2.** WF on In-the-Wild resampled to prevalence  $\pi$ , EER%/balanced accuracy%. BA has the same 50% trivial baseline in every row, where the always-spoof accuracy baseline moves from 90.00% to 99.00%. A dash marks a budget not run at that prevalence.

| $\pi$ | $N$    | Source    | $q = .02$ | $q = .10$ | $q = .30$ | BBSE      |
|-------|--------|-----------|-----------|-----------|-----------|-----------|
| 0.372 | 31,779 | 5.06/82.7 | —         | —         | 4.55/95.4 | 4.54/95.5 |
| 0.010 | 20,165 | 5.10/82.7 | 4.95/92.4 | 5.10/92.1 | 5.27/75.7 | 5.17/82.2 |
| 0.020 | 20,370 | 5.16/82.7 | 5.00/92.9 | 5.12/91.8 | 5.23/79.0 | 5.16/82.7 |
| 0.050 | 21,014 | 5.14/82.6 | 5.04/93.7 | 5.11/94.6 | 5.09/78.8 | 5.10/82.6 |
| 0.100 | 22,181 | 5.12/82.7 | 5.04/94.0 | 4.96/94.5 | 5.05/82.3 | 5.02/85.2 |
| 0.372 | 31,778 | 5.06/82.6 | —         | —         | 4.87/95.2 | —         |
| 0.900 | 13,129 | 5.26/82.7 | 5.04/93.7 | 5.26/95.0 | 6.02/84.4 | 5.03/93.7 |
| 0.950 | 12,438 | 4.66/83.6 | 4.58/92.5 | 4.94/94.6 | 5.66/83.0 | 4.52/91.6 |
| 0.970 | 12,181 | 3.74/83.5 | 3.91/93.3 | 4.66/93.7 | 5.21/81.8 | 3.84/92.3 |

determine all three unknowns, which is why a source confusion matrix may fail to transfer.

#### 4.5. Intervening on prevalence

On resampled In-the-Wild (Table 2) the same holds. At 97% spoof  $q = 0.3$  gives accuracy 65.42% against the source’s 98.94%, but balanced accuracy 81.8% against 83.5% and EER 5.21% against 3.74%;  $q = 0.02$  gains 9.8 points of balanced accuracy for 0.17 of EER.

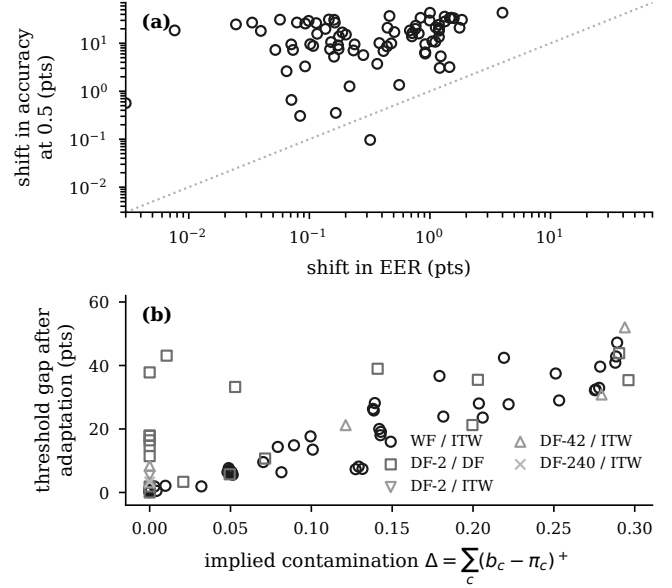
**The deployment direction.** Real pools are mostly bona fide, so we extend the sweep to  $\pi \in \{0.01, 0.02, 0.05, 0.10\}$ , where the spoof tail is the constrained one. The mirror image appears: at  $\pi = 0.01$ ,  $q = 0.3$  takes accuracy from 65.72% to 51.90% and balanced accuracy from 82.69% to 75.71%, with the damage carried entirely by false alarms (FPR 48.58% against 34.62%, FNR 0.00% in both). A small budget does not break where (3) says it must: its spoof tail is bounded at 52.7% purity and measured at 33.3%, yet reaches 97.00% accuracy. Every arm at every prevalence has EER between 4.95% and 5.27% while accuracy spans 51.90% to 97.00%.

**Budget sweeps.** Repeating three points of each sweep over three adaptation seeds shows the ordering is not seed noise: at  $\pi = 0.01$  balanced accuracy falls from  $+12.00 \pm 0.17$  at  $\Delta = 0.049$  through  $+3.35 \pm 0.62$  to  $-5.19 \pm 1.63$  at 0.289, and at  $\pi = 0.97$  from  $+11.62 \pm 0.10$  to  $-1.80 \pm 0.20$ . Each step is five to sixteen times the seed spread, and both orderings survive disjoint evaluation.

**Prior estimation in that direction.** BBSE does not transfer: it estimates 35.9–41.8% spoof on pools that are 1.1–9.7% spoof, changing accuracy by  $-1.0$  to  $+4.7$  points where a prior-free  $q = 0.02$  gains 27.1 to 31.3.

## 5. DISCUSSION AND LIMITATIONS

On these detectors, confident-tail adaptation is a way of choosing an operating point rather than of improving a detector: it moves accuracy by tens of points and EER by fractions



**Fig. 1.** Over 79 runs spanning four checkpoints, two corpora, prevalence 0.01–0.972 and budgets 0.02–0.50. (a) Shift in accuracy at threshold 0.5 against shift in EER; 78 of 79 runs lie above the diagonal, and values below  $3 \times 10^{-3}$  points are clipped to the axis. (b) The post-adaptation threshold gap against the contamination (2) implies, pooled over all strata rather than within a prevalence:  $\rho = +0.75$  here against  $+0.36$  for the budget alone.

of one. The counting obstruction is what makes the governing quantity computable, and it is useful only as a share of the pool, a purity ceiling ordering nothing. It bounds damage rather than promising a benefit; a balanced pool with a perfectly pure selection still lost 9.6 points.<sup>2</sup>

The reading we draw is that absolute pseudo-labels are the wrong instrument for the problem they are usually applied to. An objective defined on relative evidence between clips cannot be optimized by shifting all scores together, and so cannot buy an apparent gain the way a budget-driven threshold move can. The consistency term points the same way: the gain-and-noise view it enforces agreement against itself scores 9.03% EER on In-the-Wild where the unperturbed clip scores 4.91%, while a temporal crop costs nothing, so agreement is being demanded of a substantially degraded view.

The evidence is four released checkpoints from one architecture family, two corpora, an incomplete DF eval partition that rules out full-benchmark comparison, one adaptation recipe, and three seeds on eight cells, on pools that share audio within a corpus.  $\Delta$  uses realized bucket sizes and the true prior, so it is a diagnostic rather than a label-free rule, and the prior estimate one would need fails exactly where the deployment cases lie.

<sup>2</sup>Scores, audit and code: <https://github.com/RohamIzadidoost/Kaggle-deepFakeAudio>

**Ethics and assistance.** This study uses existing public datasets and released checkpoints; no new human-subject data were collected. AI coding assistants contributed to the text, code, figure and experiment scripts under the authors’ direction.

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